
Framework for Evaluating Explainability Methods in Artificial Intelligence

User Manual
XAIMetrics version 1.0
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Acronyms

XAI: Explainable Artificial Intelligence

AI: Artificial Intelligence

Introduction

The rapid advancement of Artificial Intelligence (AI) techniques has transformed numerous sectors, including healthcare, finance, cybersecurity, and logistics. However, this progress has also highlighted an important limitation: the lack of transparency of many of these models, particularly those regarded as black-box systems [1]. In many real-world contexts, obtaining accurate predictions is not sufficient; it is also essential to understand how and why a model makes certain decisions, especially when these decisions must be interpretable, auditable, or justifiable [2].

In this context, Explainable Artificial Intelligence (XAI) [3] aims to provide understandable explanations of model behaviour. Among the different approaches proposed in the literature, local explanation methods such as SHAP [4], LIME [5], *Integrated Gradients* [6], and *BreakDown* [7] have gained considerable relevance because they make it possible to analyse the contribution of input variables to specific decisions, supporting tasks such as model validation, bias detection, and increased trust in AI systems.

Despite the significant development of XAI techniques, evaluating the quality of explanations remains an open and complex problem [2]. Unlike the evaluation of models' predictive performance, for which well-established and widely accepted metrics exist, XAI does not always provide a reference explanation that can be used to directly determine the quality of an explanation. Consequently, numerous approaches and metrics have been proposed to evaluate different properties, such as fidelity, robustness, sensitivity, interpretability, and complexity [8, 9]. However, existing metrics are often implemented using *ad hoc* approaches, making them difficult to configure, compare, and reproduce across different experimental settings [10, 11]. Therefore, a unified and extensible software framework is needed to facilitate the consistent and reproducible evaluation of XAI methods.

To address these limitations, this work presents a Python library that implements 17 evaluation metrics and enables the assessment of explanations generated by different XAI methods across multiple models and datasets. The tool follows a modular and extensible architecture, allows experiments to be defined through **YAML** files, and generates comparative reports in **CSV** and **JSON** formats.

Getting and running the library

This section describes the necessary steps to download, install, and configure everything you need to use the **XAIMetrics** library.

Table 2.1 specifies the versions of dependencies required by XAIMetrics.

Dependency	Version
aix360	0.3.0
cloudpickle	3.1.2
dalex	1.8.0
interpret	0.7.8
joblib	1.5.3
lime	0.2.0.1
matplotlib	3.10.9
numpy	2.2.6 / 2.4.6
pandas	2.3.3
pyod	3.5.2
PyYAML	6.0.3
quantus	0.6.0
scikit-learn	1.7.2 / 1.8.0
seaborn	0.13.2
shap	0.49.1 / 0.50.0 / 0.51.0
torch	2.12.0

Table 2.1: Versions of dependencies used by the XAIMetrics library

The library is distributed via GitHub, available at <https://github.com/kdis-lab/XAIMetrics>. The repository contains the following main folders:

- *docs*. Contains the documentation of the library, including the user’s manual in *pdf* format and the API reference in *html*.
- *examples*. Provides a basic usage notebook, sample configuration, example hydraulic dataset files, a pretrained model, attribution files for LIME and SHAP, and generated report outputs in CSV and JSON format.

- *tests*. Contains unit tests to verify the correct functionality of the library.
- *xai_metrics*. Contains the Python source code of the library.

The library can be installed directly from the repository in editable mode:

```
$ git clone https://github.com/kdis-lab/XAIMetrics.git
$ cd XAIMetrics
$ pip install -e .
```

XAIMetrics library

XAIMetrics is a Python library that provides a unified framework for evaluating explanations generated by XAI methods. The library enables explanations to be assessed across different models, datasets, and XAI techniques under consistent experimental conditions, which allows for a systematic comparison of explanation methods.

XAIMetrics has been designed to provide a modular, extensible, and reproducible evaluation environment. It integrates metric implementation, experiment configuration, automated execution, and report generation into a single workflow. Experiments can be defined through **YAML** configuration files, allowing users to specify the datasets, models, explanations, metrics, and execution parameters without modifying the library source code.

These are the main features of the library:

- Provides an implementation of 17 metrics for evaluating different properties of XAI explanations.
- Supports the evaluation of multiple XAI methods across different models and datasets under the same experimental configuration.
- Allows multiple metrics to be executed automatically for every available combination of dataset, model, and explanation method.
- Uses declarative **YAML** configuration files to define and reuse complete evaluation experiments.
- Follows a modular and extensible architecture that facilitates the implementation and integration of new evaluation metrics.
- Supports the injection of external dependencies, including explanation, perturbation, normalization, and model-loading functions.
- Generates a single comparative report containing the experimental configuration, metric parameters, and results obtained by each XAI method.
- Exports the evaluation results in reusable **CSV** and **JSON** formats, facilitating their filtering, analysis, and integration into other environments.

In this chapter, the architecture of the library and its main functionalities are described. Moreover, the complete evaluation workflow is presented, including experiment configuration, evaluation-context construction, metric executions, and report generation

3.1 XAIMetrics library architecture

The library has been developed as an open-source Python project. It follows a modular architecture that separates the different responsibilities involved in the evaluation process. This organization makes it possible to extend the library with new metrics or components without modifying its main execution workflow.

The library is organized into five main modules: `base`, `config`, `metrics`, `runner`, and `reporting`. The main functionality of each module is described below:

- *base*. Contains the common abstractions and utilities used throughout the library. It defines the `BaseMetric` class, from which all evaluation metrics inherit, and the `MetricContext` class, which groups the information required to execute a metric, including the evaluated model, test data, labels, selected observations, feature attributions, and execution device. This module also provides the metric registration mechanism through the `register_metric` decorator, allowing new metrics to be discovered and instantiated automatically.
- *config*. Manages the experimental configuration. It is responsible for loading, interpreting, and validating the information required to define an evaluation experiment. From a YAML file or a configuration provided at runtime, this module identifies the paths to datasets, models, and attributions, as well as the selected metrics, their parameters, and the execution device. This separation allows experiments to be modified and reused without changing any code.
- *runner*. Coordinates the execution of the evaluation process. It discovers the registered metrics, selects those specified in the experimental configuration, constructs the required `MetricContext` objects, and executes each metric for every available combination of dataset, model, and XAI method. It also supports external dependencies provided at runtime, such as explanation, perturbation, normalization, and model-loading functions.
- *reporting*. Organizes and exports the results obtained during the evaluation. It aggregates the individual metric values and collects all the relevant experimental information in a single comparative report. This report includes the datasets, models, evaluated metrics, metric parameters, XAI methods, and corresponding results. The resulting report can be exported in CSV and JSON formats for subsequent analysis and reuse.
- *metrics*. Contains the implementation of the evaluation metrics included in the library. Each metric is implemented as a class that inherits from `BaseMetric`, receives a `MetricContext`, and defines its specific computation through the `run()` method. The metrics are organized into submodules according to the main explainability property they evaluate, keeping the implementation of each metric independent and facilitating the addition of new measures.
- *metrics.complexity*. Contains the implementation of Complexity Metrics.
- *metrics.faithfulness*. Contains the implementation of Faithfulness Metrics.
- *metrics.robustness*. Contains the implementation of Robustness Metrics.
- *metrics.sensitivity*. Contains the implementation of Sensitivity Metrics.
- *metrics.fidelity*. Contains the implementation of Fidelity Metrics.
- *metrics.fidelity.completeness*. Contains the implementation of Completeness Metrics.
- *metrics.fidelity.soundness*. Contains the implementation of Soundness Metrics.

Figure 3.1 illustrates the overall architecture of the **XAIMetrics** library, showing its main modules and the relationships between the components involved in experiment configuration, metric execution, and report generation.

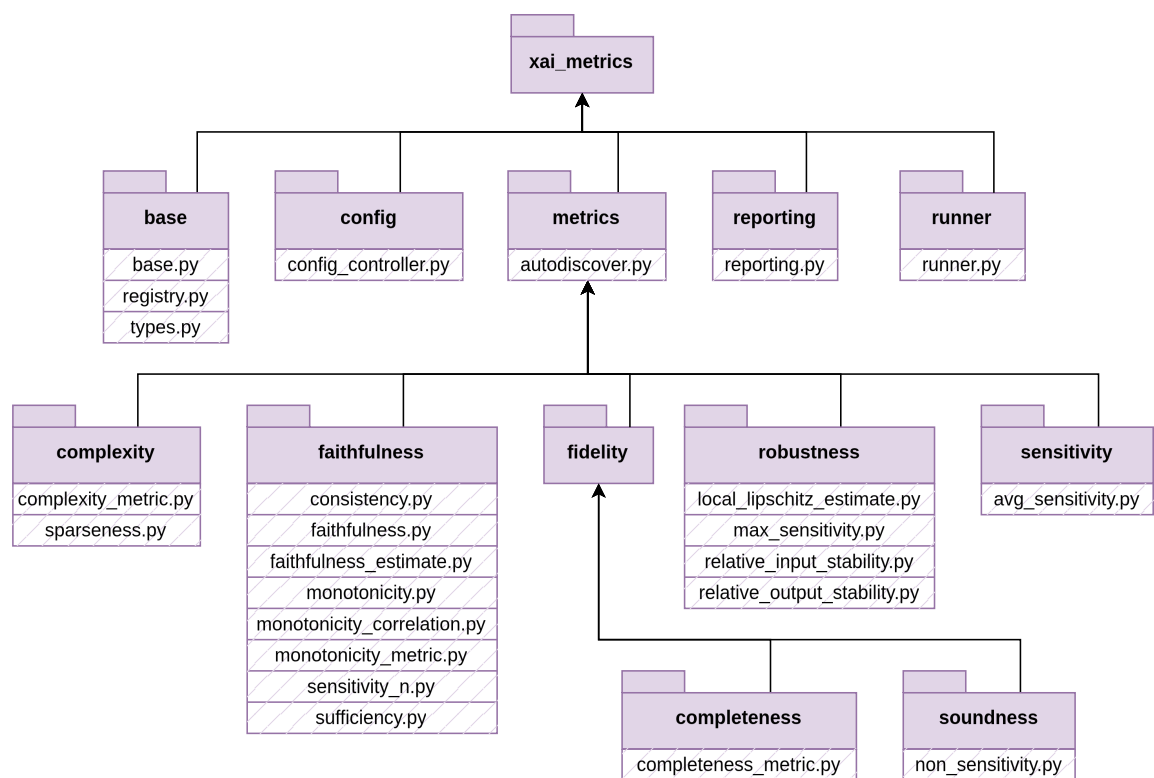


Figure 3.1: Architecture XAIMetrics library

Run available Metrics in the Library

In this chapter, the individual metrics implemented in the **XAIMetrics** library are presented. For each metric, a brief description of its purpose and the explainability property it evaluates is provided, followed by two illustrative examples. The first example shows how to execute the metric individually using its corresponding Python class, whereas the second demonstrates how to include it in an automated evaluation experiment configured through a **YAML** file and executed by the **runner** module

The metrics are organized according to the objective evaluation categories introduced in the proposed taxonomy and currently supported by the library. These include Faithfulness in Section 4.1, Sensitivity in Section 4.3, Robustness in Section 4.4, Complexity in Section 4.2, and Fidelity ???. The latter is further divided into Completeness in Section ?? and Soundness in Section ??.

4.1 Faithfulness

This category includes metrics that evaluate the extent to which an explanation reflects the predictive behaviour of the model in terms of feature contributions. For attribution-based explanations, these metrics analyse whether the features identified as the most relevant are also those that have the greatest influence on the model output. Therefore, they assess the relationship between the importance assigned by the explanation and the actual effect of each feature on the prediction.

Faithfulness metrics are particularly useful for determining whether an explanation accurately represents the decision-making process of the model. They make it possible to verify whether modifying, removing, or introducing the most relevant features produces the expected changes in the model output. In this way, they provide a quantitative assessment of whether the generated attributions are consistent with the model behaviour.

However, the results of these metrics may depend on the perturbation strategy, the feature ordering, and the presence of dependencies between input variables. Modifying features independently may generate unrealistic samples or fail to preserve the relationships present in the original data. Consequently, faithfulness scores should be interpreted while considering the characteristics of the dataset and the experimental configuration used to evaluate the explanations.

4.1.1 Consistency

The Consistency metric evaluates whether observations with similar explanations are assigned to the same predicted class [12]. To perform this comparison, the continuous attribution vectors are first transformed into discrete representations. For each observation, the metric computes the proportion of the remaining observations in its group that receive the same predicted class.

A high consistency score indicates that observations with equivalent attribution patterns frequently receive the same model prediction. Conversely, a low score indicates that similar explanations may be associated with different predicted classes. Therefore, the metric assesses whether the explanations consistently represent the decision-making behaviour of the model.

The parameters used by this metric can be summarized as follows:

- **abs** (bool): whether to use the absolute values of the attributions before computing the metric. If all attribution values are negative and this parameter is enabled, their absolute values are used. Otherwise, the metric is skipped. (*Default: True*)
- **normalise** (bool): whether to normalise the attribution values before discretising and comparing the explanations. (*Default: True*)

The implementation uses the default discretisation and normalisation functions provided by **Quantus**. In particular, the default discretisation function is `top_n_sign`, which converts each continuous attribution vector into a discrete representation according to the most relevant features and the signs of their attributions.

Before computing the metric, the model is set to evaluation mode, and the observations specified in the **MetricContext** are selected together with their labels and attribution values. The metric returns a consistency score for each evaluated observation. When all attribution values are negative, their treatment depends on the **abs** parameter. If **abs=True**, their absolute values are used and the evaluation continues. If **abs=False**, the metric raises a **MetricSkipped** exception and is not evaluated for that attribution configuration.

The metric is implemented through the **Consistency** class, placed in the module `xai_metrics.metrics.faithfulness`. It can be executed directly using this class (Section 4.1.1.1), or as part of the automated evaluation pipeline provided by `run_evaluation` (Section 4.1.1.2).

Both usage modes rely on the same **YAML** configuration file. The experiment evaluates explanations generated by **LIME** for an **Isolation Forest** model on the hydraulic dataset. The configuration also specifies that the original signs of the attribution values must be preserved by setting **abs** to **false**, while attribution normalisation remains enabled.

Examples of these usage modes are available in the library under: `examples/specific_examples/Consistency_examples.py`.

4.1.1.1 Direct evaluation using the Consistency class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `Consistency` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import Consistency
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = Consistency(
7     context=context,
8     params={
9         "abs": False,
10        "normalise": True
11    }
12 )
13 scores = metric.run()

```

4.1.1.2 Automated evaluation using run_evaluation

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the YAML configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered "Consistency" metric, although this argument would not be necessary if `Consistency` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["Consistency"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset_name: "hydraulic"
  model_name: "IForest"
  xai_method_name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"
  attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:

```

```
- name: Consistency
  params:
    abs: false
    normalise: true
```

4.1.2 Faithfulness Estimate

The Faithfulness Estimate metric evaluates whether the importance assigned to the input features by an explanation is consistent with their actual influence on the model output [11]. To perform this evaluation, features are perturbed in groups according to their attribution importance. For each perturbation step, the metric compared the sum of the attribution values of the perturbed features with the corresponding decrease in the target model output.

The agreement between these two sequences is measured using the Pearson correlation coefficient. A high positive score indicates that features with larger attribution values also produce larger changes in the model output when perturbed. Conversely, a low or negative score indicates weak or inverse agreement between the feature importance assigned by the explanation and the behaviour of the model. Therefore, the metric assesses whether the attribution values accurately reflect the influence of the features on the prediction.

The parameters used by this metric can be summarized as follows:

- **features_in_step** (int): number of features perturbed together at each evaluation step. (*Default: 1*)
- **abs** (bool): whether to use the absolute values of the attributions before computing the metric. If all attribution values are negative and this parameter is enabled, their absolute values are used. Otherwise, the metric is skipped. (*Default: False*)
- **normalise** (bool): whether to normalise the attribution values before computing the metric. (*Default: True*)
- **perturb_baseline** (str): baseline used to replace the selected features during perturbation. The available values depend on the perturbation function provided by **Quantus** and commonly include "black", "white", "mean", "random", and "uniform". (*Default: "black"*)

The implementation uses the default normalisation and perturbation functions provided by **Quantus**. The similarity between the attribution sums and the corresponding decreases in the model output is calculated using an internal safe implementation of the Pearson correlation coefficient. When either of the compared vectors has zero variance, this function returns 0.0 instead of an undefined NaN value.

Before computing the metric, the model is set to evaluation mode, and the observations specified in the **MetricContext** are selected together with their labels and attribution values. The metric returns one Faithfulness Estimate score for each evaluated observation. Higher values indicate stronger agreement between the importance assigned to the features and the changes produced in the model output after perturbation. When all attribution values are negative, their treatment depends on the **abs** parameter. If **abs=True**, their absolute values are used, and the evaluation continues. If **abs=False**, the metric raises a **MetricSkipped** exception and is not evaluated for that attribution configuration.

The metric is implemented through the `FaithfulnessEstimate` class in the module `xai_metrics.metrics.faithfulness`. It can be executed directly using this class (Section 4.1.2.1) or as part of the automated evaluation pipeline provided by `run_evaluation` (Section 4.1.2.2).

Both usage modes rely on the same YAML configuration file. The experiment evaluates explanations generated by LIME for an Isolation Forest model on the hydraulic dataset. The configuration perturbs one feature at each step, preserves the original signs of the attribution normalisation, and uses the mean feature value as the perturbation baseline.

Examples of both usage modes are available in the library under: `examples/specific_examples/FaithfulnessEstimate_examples.py`

4.1.2.1 Direct evaluation using the `FaithfulnessEstimate` class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `FaithfulnessEstimate` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import FaithfulnessEstimate
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = FaithfulnessEstimate(
7     context=context,
8     params={
9         "features_in_step": 1,
10        "abs": False,
11        "normalise": True,
12        "perturb_baseline": "mean"
13    }
14 )
15 scores = metric.run()

```

4.1.2.2 Automated evaluation using `run_evaluation`

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the YAML configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered "`FaithfulnessEstimate`" metric, although this argument would not be necessary if `FaithfulnessEstimate` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["FaithfulnessEstimate"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```
context:
  dataset__name: "hydraulic"
  model__name: "IForest"
  xai__method__name: "LIME"
  device: "cpu"
  model__path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X__test__path: "data/hydraulic/X_test.csv"
  y__test__path: "data/hydraulic/y_test.csv"
  attributions__path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: FaithfulnessEstimate
    params:
      features__in__step: 1
      abs: false
      normalise: true
      perturb__baseline: mean
```

4.1.3 Faithfulness

The Faithfulness metric evaluates whether the attribution values assigned to the input features reflect their actual influence on the model prediction [11]. For each evaluated observation, the metric first determines the class predicted by the model. It then replaces each feature individually with its corresponding baseline value and records the probability assigned to the originally predicted class after each replacement.

The score is computed as the negative Pearson correlation between the attribution values and the predicted-class probabilities obtained after replacing the individual features. Features with larger attribution values are expected to produce larger decreases in the predicted-class probability when replaced by their baseline values. Therefore, a high positive score indicates strong agreement between the importance assigned by the explanation and the influence of the features on the model prediction. Conversely, a low or negative score indicates weak or inverse agreement.

The parameters used by this metric can be summarized as follows:

- **base_values** (array-like, optional): explicit baseline values used to replace the input features. The provided array must contain one baseline value for each feature. When specified, these values take priority over **base_strategy**. (*Default: None*)
- **base_strategy** (str): strategy used to calculate a common baseline vector from the complete test dataset when **base_values** is not provided. The supported strategies are:
 - "mean": uses the feature-wise mean of the test dataset;
 - "median": uses the feature-wise median of the test dataset;
 - "zero": uses a vector containing one zero for each feature.

(*Default: "mean"*)

Before computing the metric, a common baseline vector is obtained either from the values explicitly provided through `base_values` or from the complete test dataset using the selected `base_strategy`. The observations specified in the `MetricContext` are then selected and evaluated independently together with their corresponding attribution vectors.

The metric returns one Faithfulness score for each evaluated observation. Higher positive values indicate stronger agreement between the attribution values and the decrease in predicted-class probability caused by replacing individual features. If the attribution vector or the probabilities obtained from the perturbed inputs have zero variance, the Pearson correlation is undefined, and the underlying AIX360 implementation may return NaN.

If an unsupported value is provided through `base_strategy`, the metric raises a `ValueError`. Moreover, the implementation relies on the `faithfulness_metric` function provided by AIX360, which required a classification model exposing a `predict_proba` method. If the evaluated model does not implement this method, the metric raises an `AttributeError`.

The metric is implemented through the `Faithfulness` class, placed in the module `xai_metrics.metrics.faithfulness`. It can be executed directly using this class (Section 4.1.3.1) or as part of the automated evaluation pipeline provided by `run_evaluation` (Section 4.1.3.2).

Both usage modes rely on the same YAML configuration file. The experiment evaluates explanations generated by LIME for an Isolation Forest model on the hydraulic dataset. The baseline vector is calculated using the feature-wise mean of the complete test dataset. This experiment assumes that the loaded model provides the `predict_proba` method required by the metric.

Examples of both usage modes are available in the library under: `examples/specific_examples/Faithfulness_examples.py`

4.1.3.1 Direct evaluation using the Faithfulness class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `Faithfulness` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import Faithfulness
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = Faithfulness(
7     context=context,
8     params={
9         "base_strategy": "mean"
10    }
11 )
12 scores = metric.run()
```

4.1.3.2 Automated evaluation using `run_evaluation`

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the **YAML** configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered **"Faithfulness"** metric, although this argument would not be necessary if **Faithfulness** is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["Faithfulness"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset_name: "hydraulic"
  model_name: "IForest"
  xai_method_name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"
  attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: Faithfulness
    params:
      base_strategy: mean

```

4.1.4 Monotonicity Correlation

The Monotonicity Correlation metric evaluates whether feature attribution values are monotonically related to the uncertainty produced in the model output when the corresponding features are perturbed [13]. For each evaluated observation, the features are ordered according to their attribution values and divided into groups. Each group is perturbed repeatedly, and the uncertainty associated with that group is estimated from the changes produced in the target model output.

The metric compares the sum of the attribution values in each feature group with the corresponding uncertainty estimate using the Spearman rank correlation coefficient. A high positive score indicates that groups containing features with greater attribution values tend to produce greater uncertainty in the model output when perturbed. Conversely, a low or negative score indicates a weak or inverse monotonic relationship between the importance assigned by the explanation and the effect of perturbing the corresponding features. Therefore, the metric assesses whether the ordering induced by the attribution values agrees with the sensitivity of the model output to feature perturbations.

The parameters used by this metric can be summarized as follows:

- **eps (float)**: threshold used when calculating the inverse prediction factor employed to scale the changes in the model output. This parameter avoids numerical instability when the magnitude of the original output is close to zero. (*Default: 1e-5*)
- **nr_samples (int)**: number of perturbed samples generated for each feature group to estimate the uncertainty associated with its perturbation. (*Default: 100*)
- **features_in_step (int)**: number of features included in each group and perturbed together at every evaluation step. (*Default: 1*)
- **abs (bool)**: whether to use the absolute values of the attributions before computing the metric. If all attribution values are negative and this parameter is enabled, their absolute values are used. Otherwise, the metric is skipped. (*Default: True*)
- **normalise (bool)**: whether to normalise the attribution values before ordering the features and computing the metric. (*Default: True*)
- **perturb_baseline (str)**: baseline used when perturbing the selected features. The available values depend on the perturbation function provided by **Quantus** and commonly include "black", "white", "mean", "random", and "uniform". (*Default: "uniform"*)

The implementation uses the default normalisation and perturbation functions provided by **Quantus**. The relationship between the attribution sums and the uncertainty estimates is calculated using an internal safe implementation of the Spearman correlation coefficient. When either of the compared vectors has zero variance, this function returns 0.0 instead of an undefined NaN value.

Before computing the metric, the model is set to evaluation mode, and the observations specified in the **MetricContext** are selected together with their target labels and attribution values. For each observation, the feature groups are perturbed **nr_samples** times. The uncertainty associated with each group is estimated from the mean squared change in the target model output, scaled relative to the magnitude of the original output. The metric then returns one Monotonicity Correlation score for each evaluated observation.

Higher values indicate a stronger positive monotonic relationship between feature importance and the uncertainty caused by perturbing the corresponding feature groups. When all attribution values are negative, their treatment depends on the **abs** parameter. If **abs=True**, their absolute values are used and the evaluation continues. If **abs=False**, the metric raises a **MetricSkipped** exception and is not evaluated for that attribution configuration.

The metric is implemented through the **MonotonicityCorrelation** class in the module **xai_metrics.metrics.faithfulness**. It can be executed directly using this class (Section 4.1.4.1) or as part of the automated evaluation pipeline provided by **run_evaluation** (Section 4.1.4.2).

Both usage modes rely on the same YAML configuration file. The experiment evaluates explanations generated by LIME for an Isolation Forest model on the hydraulic dataset. The configuration perturbs one feature at each step, generates 100 perturbation samples for each feature group, uses the absolute and normalised attribution values, and replaces the perturbed features with their mean values.

Examples of both usage modes are available in the library under: **examples/specific_examples/MonotonicityCorrelation_examples.py**

4.1.4.1 Direct evaluation using the MonotonicityCorrelation class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `MonotonicityCorrelation` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import MonotonicityCorrelation
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = MonotonicityCorrelation(
7     context=context,
8     params={
9         "eps": 1e-5,
10        "nr_samples": 100,
11        "features_in_step": 1,
12        "abs": True,
13        "normalise": True,
14        "perturb_baseline": "mean"
15    }
16 )
17 scores = metric.run()

```

4.1.4.2 Automated evaluation using run_evaluation

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the YAML configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered `"MonotonicityCorrelation"` metric, although this argument would not be necessary if `MonotonicityCorrelation` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["MonotonicityCorrelation"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset__name: "hydraulic"
  model__name: "IForest"
  xai_method__name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"

```



```

attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
- name: MonotonicityCorrelation
  params:
    eps: 1e-5
    nr_samples: 100
    features_in_step: 1
    abs: true
    normalise: true
    perturb_baseline: mean

```

4.1.5 Monotonicity Metric

The Monotonicity Metric evaluates whether the probability assigned to the originally predicted class increases monotonically as the input features are progressively restored from a baseline vector [14]. For each evaluated observation, the metric first obtains the class predicted by the model using the complete original input. It then creates an initial input containing the baseline value of each feature and progressively restores the original feature values in increasing order of their attribution values.

After restoring each feature, the metric records the probability assigned to the originally predicted class. The resulting sequence is expected to be monotonically non-decreasing, meaning that restoring progressively more relevant information should not reduce the probability of the original prediction. The metric returns **True** when this condition is satisfied and **False** otherwise.

The parameters used by this metric can be summarized as follows:

- **base_values** (array-like, optional): explicit baseline values used as the initial feature values. The provided array must contain one baseline value for each input feature. When specified, these values take priority over **base_strategy**. (*Default: None*)
- **base_strategy** (str): strategy used to calculate a common baseline vector from the complete test dataset when **base_values** is not provided. The supported strategies are:
 - "mean": uses the feature-wise mean of the test dataset;
 - "median": uses the feature-wise median of the test dataset;
 - "zero": uses a vector containing one zero for each feature.

(*Default: "mean"*)

Before computing the metric, a common baseline vector is obtained either from the values explicitly provided through **base_values** or from the complete test dataset using the selected **base_strategy**. The observations specified in the **MetricContext** are then selected and evaluated independently together with their corresponding attribution vectors.

For each observation, the features are ordered by increasing attribution value. Starting from the baseline vector, their original values are restored cumulatively following this order. After each restoration, the probability assigned to the class predicted for the original complete input is obtained.

The metric returns one Boolean result for each evaluated observation. A value of **True** indicates that the predicted-class probability never decreases as the features are progressively restored,

whereas `False` indicates that the monotonicity condition is violated. When the Boolean results are averaged, `True` and `False` are interpreted as (1) and (0), respectively. Therefore, the resulting mean represents the proportion of evaluated observations that satisfy the monotonicity criterion.

If an unsupported value is provided through `base_strategy`, the metric raises a `ValueError`. Moreover, the implementation relies on the `monotonicity_metric` function provided by AIX360, which requires a classification model exposing a `predict_proba` method. If the evaluated model does not implement this method, the metric raises an `AttributeError`.

The metric is implemented through the `MonotonicityMetric` class in the module `xai_metrics.metrics.faithfulness`. It can be executed directly using this class (Section 4.1.5.1) or as part of the automated evaluation pipeline provided by `run_evaluation` (Section 4.1.5.2).

Both usage modes rely on the same YAML configuration file. The experiment evaluates explanations generated by LIME for an Isolation Forest model on the hydraulic dataset. The initial baseline vector is calculated using the feature-wise mean of the complete test dataset. The experiment assumes that the loaded model provides the `predict_proba` method required by the metric.

Examples of both usage modes are available in the library under: `examples/specific_examples/MonotonicityMetric_examples.py`

4.1.5.1 Direct evaluation using the `MonotonicityMetric` class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `MonotonicityMetric` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import MonotonicityMetric
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = MonotonicityMetric(
7     context=context,
8     params={
9         "base_strategy": "mean"
10    }
11 )
12 scores = metric.run()

```

4.1.5.2 Automated evaluation using `run_evaluation`

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the YAML configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered "`MonotonicityMetric`" metric, although this argument would not be necessary if `MonotonicityMetric` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["MonotonicityMetric"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset_name: "hydraulic"
  model_name: "IForest"
  xai_method_name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"
  attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: MonotonicityMetric
    params:
      base_strategy: mean

```

4.1.6 Monotonicity

The Monotonicity metric evaluates whether the target model output evolves monotonically as feature groups are progressively introduced according to their attribution values [14]. For each evaluated observation, the features are ordered by increasing attribution value. The evaluation starts from an input defined by a baseline, and the original feature values are progressively introduced in groups following this ordering.

After introducing each feature group, the metric evaluates the model output associated with the target label. The resulting sequence is expected to be monotonically non-decreasing, meaning that adding progressively more relevant information should not reduce the target model output. The metric returns **True** when this condition is satisfied throughout the complete sequence and **False** when the output decreases in at least one step.

The parameters used by this metric can be summarized as follows:

- **features_in_step** (int): number of features introduced together at each evaluation step. (*Default: 1*)
- **abs** (bool): whether to use the absolute values of the attributions before ordering the features and computing the metric. If all attribution values are negative and this parameter is enabled, their absolute values are used. Otherwise, the metric is skipped. (*Default: True*)
- **normalise** (bool): whether to normalise the attribution values before ordering the features and computing the metric. (*Default: True*)

- **perturb_baseline** (str): baseline used to initialise the input before the original feature values are progressively introduced. The available values depend on the perturbation function provided by **Quantus** and commonly include "black", "white", "mean", "random", and "uniform". (*Default*: "black")

The implementation uses the default normalisation and perturbation functions provided by **Quantus**. For each observation, the input is initially replaced by the selected baseline. The original feature values are then introduced in groups of size **features_in_step**, following the increasing order of their attribution values.

Before computing the metric, the model is set to evaluation mode, and the observations specified in the **MetricContext** are selected together with their target labels and attribution values. After each feature group is introduced, the target model output is evaluated and compared with the output obtained in the previous step.

The metric returns one Boolean value for each evaluated observation. A value of **True** indicates that the target model output never decreases as the feature groups are progressively introduced, whereas **False** indicates that the monotonicity condition is violated at least once. When the Boolean results are averaged, **True** and **False** are interpreted as (1) and (0), respectively. Therefore, the resulting mean represents the proportion of observations that satisfy the monotonicity criterion.

When all attribution values are negative, their treatment depends on the **abs** parameter. If **abs=True**, their absolute values are used and the evaluation continues. If **abs=False**, the metric raises a **MetricSkipped** exception and is not evaluated for that attribution configuration.

The metric is implemented through the **Monotonicity** class, placed in the module **xai_metrics.metrics.faithfulness**. It can be executed directly using this class (Section 4.1.6.1) or as part of the automated evaluation pipeline provided by **run_evaluation** (Section 4.1.6.2).

Both usage modes rely on the same **YAML** configuration file. The experiment evaluates explanations generated by **LIME** for an Isolation Forest model on the hydraulic dataset. The configuration introduces one feature at each step, uses absolute and normalised attribution values, and initialises the perturbed input using the "black" baseline.

Examples of both usage modes are available in the library under: **examples/specific_examples/Monotonicity_examples.py**

4.1.6.1 Direct evaluation using the Monotonicity class

In the direct evaluation mode, the experimental context is constructed from the **config.yaml** file using the **ConfigController** class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the **Monotonicity** class is instantiated directly using the desired parameters. Calling the **run()** method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import Monotonicity
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = Monotonicity(
7     context=context,
8     params={

```

```

9         "features_in_step": 1,
10        "abs": True,
11        "normalise": True,
12        "perturb_baseline": "black"
13    }
14 )
15 scores = metric.run()

```

4.1.6.2 Automated evaluation using run_evaluation

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the YAML configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered "Monotonicity" metric, although this argument would not be necessary if `Monotonicity` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["Monotonicity"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset_name: "hydraulic"
  model_name: "IForest"
  xai_method_name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"
  attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: Monotonicity
    params:
      features_in_step: 1
      abs: true
      normalise: true
      perturb_baseline: black

```

4.1.7 Sensitivity-N

The Sensitivity-N metric evaluates whether the attribution values assigned to groups of input features agree with the changes produced in the target model output when those features are perturbed [14]. For each evaluated observation, the features are ordered by decreasing attribution

value and progressively perturbed in groups. At each step, the metric records both the attribution values associated with the processed feature group and the corresponding variation in the target model output.

The agreement between these quantities is measured using the Pearson correlation coefficient. A high positive correlation indicates that feature groups with greater attribution values tend to produce larger changes in the target output when perturbed. Conversely, a low or negative correlation indicates weak or inverse agreement between the importance assigned by the explanation and the actual influence of the corresponding features on the model prediction. Therefore, the metric assesses whether the attribution values accurately represent the sensitivity of the model output to feature perturbations.

The parameters used by this metric can be summarized as follows:

- **n_max_percentage** (float): maximum proportion of input features considered during the evaluation. For example, a value of 0.8 limits the experiment to perturbation steps covering up to 80% of the available features. (*Default: 0.8*)
- **features_in_step** (int): number of features perturbed together at each evaluation step. (*Default: 1*)
- **abs** (bool): whether to use the absolute values of the attributions before ordering the features and computing the metric. If all attribution values are negative and this parameter is enabled, their absolute values are used. Otherwise, the metric is skipped. (*Default: False*)
- **normalise** (bool): whether to normalise the attribution values before computing the metric. (*Default: True*)
- **perturb_baseline** (str): baseline used to replace the selected features during perturbation. The available values depend on the perturbation function provided by **Quantus** and commonly include "black", "white", "mean", "random", and "uniform". (*Default: "black"*)

The implementation uses the default Pearson correlation, normalisation function, and baseline-replacement function provided by **Quantus**. For each observation, the features are ordered by decreasing attribution value and processed in groups of size **features_in_step**. Only the perturbation steps covered by **n_max_percentage** are included in the evaluation.

Before computing the metric, the model is set to evaluation mode, and the observations specified in the **MetricContext** are selected together with their target labels and attribution values. After each feature group is perturbed, the variation in the target model output is compared with the attribution values associated with that group.

The metric returns the Sensitivity-N result produced by **Quantus**. Higher values indicate stronger agreement between the attribution values and the changes caused in the target model output by perturbing the corresponding features. Since result aggregation is enabled by default in the underlying **Quantus** implementation, the returned values may represent an aggregation of the correlations calculated across the evaluated perturbation steps rather than one independent score for every input observation.

When all attribution values are negative, their treatment depends on the **abs** parameter. If **abs=True**, their absolute values are used and the evaluation continues. If **abs=False**, the metric raises a **MetricSkipped** exception and is not evaluated for that attribution configuration.

The metric is implemented through the **SensitivityN** class, placed in the module **xai_metrics.metrics.faithfulness**. It can be executed directly using this class (Section 4.1.7.1) or as part of the automated evaluation pipeline provided by **run_evaluation** (Section 4.1.7.2).

Both usage modes rely on the same YAML configuration file. The experiment evaluates explanations generated by LIME for an Isolation Forest model on the hydraulic dataset. In this example, all input features are considered, one feature is perturbed at each step, the original signs of the attribution values are preserved, normalisation is enabled, and the perturbed features are replaced using the "uniform" baseline.

Examples of both usage modes are available in the library under: `examples/specific_examples/SensitivityN_examples.py`

4.1.7.1 Direct evaluation using the SensitivityN class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `SensitivityN` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import SensitivityN
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = SensitivityN(
7     context=context,
8     params={
9         "n_max_percentage": 1.0,
10        "features_in_step": 1,
11        "abs": False,
12        "normalise": True,
13        "perturb_baseline": "uniform"
14    }
15 )
16 scores = metric.run()

```

4.1.7.2 Automated evaluation using run_evaluation

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the YAML configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered "SensitivityN" metric, although this argument would not be necessary if `SensitivityN` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["SensitivityN"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset__name: "hydraulic"
  model__name: "IForest"
  xai__method__name: "LIME"
  device: "cpu"
  model__path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X__test__path: "data/hydraulic/X_test.csv"
  y__test__path: "data/hydraulic/y_test.csv"
  attributions__path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: SensitivityN
    params:
      n__max__percentage: 1.0
      features__in__step: 1
      abs: false
      normalise: true
      perturb__baseline: uniform

```

4.1.8 Sufficiency

The Sufficiency metric evaluates whether observations with similar explanations receive the same model prediction [12]. To perform this evaluation, the attribution vectors of all observations included in the evaluation batch are flattened and compared using a pairwise distance function. The resulting distance matrix is normalised, and two explanations are considered similar when the distance between them is less than or equal to a predefined threshold.

For each evaluated observation, the metric identifies the remaining observations whose explanations satisfy this similarity condition and calculates the proportion that receive the same predicted class. The observation itself is excluded from the comparison. A high score indicates that observations with similar explanations frequently exhibit the same model behaviour. Conversely, a low score indicates that similar attribution patterns may be associated with different predicted classes. If no other explanation lies within the specified distance threshold, the metric returns a score of 0.0 for that observation.

The parameters used by this metric can be summarized as follows:

- **threshold** (float): maximum normalised distance between two attribution vectors for their explanations to be considered similar. Lower values impose a stricter similarity criterion, whereas higher values allow more distant explanations to be included in the comparison. (*Default: 0.6*)
- **distance_func** (str): distance function used to compare the attribution vectors. The value is passed to Quantus and is generally a valid distance name supported by SciPy, such as "seuclidean". (*Default: "seuclidean"*)
- **abs** (bool): whether to use the absolute values of the attributions before computing the distances between explanations. If all attribution values are negative and this parameter is enabled, their absolute values are used. Otherwise, the metric is skipped. (*Default: True*)
- **normalise** (bool): whether to normalise the attribution values before comparing the explanations. (*Default: True*)

The implementation uses the default attribution normalisation function provided by `Quantus`. For the complete set of evaluated observations, the attribution vectors are flattened and their pairwise distances are calculated using `distance_func`. After normalising the distance matrix, the `threshold` parameter is used to determine which explanations are sufficiently similar.

Before computing the metric, the model is set to evaluation mode, and the observations specified in the `MetricContext` are selected together with their labels and attribution values. The model predictions are then compared among observations whose explanations satisfy the similarity criterion.

The metric returns one Sufficiency score for each evaluated observation. Scores range from 0.0 to 1.0. Higher values indicate that observations with similar explanations more frequently receive the same predicted class. Since each explanation is compared only with the other observations included in the same evaluation batch, the resulting scores depend on the observations selected in the `MetricContext`. Changing the evaluated batch may therefore change the neighbourhood of each explanation and its resulting score.

When all attribution values are negative, their treatment depends on the `abs` parameter. If `abs=True`, their absolute values are used and the evaluation continues. If `abs=False`, the metric raises a `MetricSkipped` exception and is not evaluated for that attribution configuration.

The metric is implemented through the `Sufficiency` class, placed in the module `xai_metrics.metrics.faithfulness`. It can be executed directly using this class (Section 4.1.8.1) or as part of the automated evaluation pipeline provided by `run_evaluation` (Section 4.1.8.2).

Both usage modes rely on the same YAML configuration file. The experiment evaluates explanations generated by LIME for an Isolation Forest model on the hydraulic dataset. In this example, all input features are considered, one feature is perturbed at each step, the original signs of the attribution values are preserved, normalisation is enabled, and the perturbed features are replaced using the "uniform" baseline.

Examples of both usage modes are available in the library under: `examples/specific_examples/Sufficiency_examples.py`

4.1.8.1 Direct evaluation using the Sufficiency class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `Sufficiency` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import Sufficiency
3
4 config_path = "examples/specific_examples/config.yaml"
5 metric = Sufficiency(
6     context=context,
7     params={
8         "threshold": 0.6,
9         "distance_func": "seuclidean",
10        "abs": True,
11        "normalise": True
12    }
13 )

```

4.1.8.2 Automated evaluation using `run_evaluation`

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the YAML configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered "Sufficiency" metric, although this argument would not be necessary if `Sufficiency` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["Sufficiency"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset_name: "hydraulic"
  model_name: "IForest"
  xai_method_name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"
  attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: Sufficiency
    params:
      threshold: 0.6
      distance_func: seuclidean
      abs: true
      normalise: true

```

4.2 Complexity

This category includes metrics that evaluate the extent to which an explanation is concise and concentrated on a limited number of input features or explanatory elements [15]. For attribution-based explanations, these metrics analyse how the attribution values are distributed across the input features. An explanation is considered less complex when most of its relevance is assigned to a small subset of features, whereas explanations that distribute importance across many features are considered more complex.

Complexity metrics are particularly useful for assessing whether an explanation can be represented in a compact and understandable form. Explanations involving fewer relevant features may be easier to inspect and interpret, especially in high-dimensional problems. These metrics can quantify properties such as the entropy or dispersion of the attribution distribution, as well as the degree to which feature importance is concentrated on a small number of variables.

However, a less complex explanation is not necessarily more faithful or informative. Excessively sparse explanations may omit features that contribute meaningfully to the model prediction, while more distributed explanations may accurately reflect models whose decisions depend on multiple interacting variables. Consequently, complexity scores should be interpreted together with other evaluation properties and considering the characteristics of the model, dataset, and explanation method.

4.2.1 Complexity

The Complexity metric evaluates how widely the attribution importance is distributed across the input features [10]. For each evaluated observation, the absolute attribution values are converted into relative contributions by dividing the importance of each feature by the total attribution distribution.

An explanation whose importance is concentrated on a small number of features produces a low entropy value and is therefore considered less complex. Conversely, when the attribution importance is distributed more uniformly across many features, the entropy increases and the explanation is considered more complex. Thus, lower scores indicate more concise explanations, whereas higher scores indicate that a larger number of features contribute substantially to the explanation.

The parameters used by this metric can be summarized as follows:

- **normalise (bool)**: whether to normalise the attribution values before computing their relative contribution and entropy. When this parameter is enabled, the default normalisation function provided by **Quantus** is used. (*Default: True*)

The absolute-value operation is always applied to the attribution values before computing the metric. This behaviour is defined by the underlying **Quantus** implementation and is fixed in the wrapper since the **abs** parameter is not exposed through the metric configuration. In addition, if all the attribution values supplied to the wrapper are negative, they are explicitly converted to their absolute values before being passed to **Quantus**.

Before computing the metric, the model is set to training mode, and the observations specified in the **MetricContext** are selected together with their labels and attribution values. The metric returns one Complexity score for each evaluated observation. Lower values indicate that the attribution importance is concentrated on fewer features, while higher values indicate a more widely distributed attribution pattern. The mean of the returned scores can be used as an aggregated measure of explanation complexity for the evaluated observations.

The metric is implemented through the **Complexity** class, which can be found in the module **xai_metrics.metrics.complexity**. It can be executed directly using this class (Section 4.2.1.1) or as part of the automated evaluation pipeline provided by **run_evaluation** (Section 4.2.1.2).

Both usage modes rely on the same **YAML** configuration file. The experiment evaluates explanations generated by **LIME** for an Isolation Forest model on the hydraulic dataset. Attribution normalisation is enabled before calculating the entropy-based complexity score.

Examples of both usage modes are available in the library under: **examples/specific_examples/Complexity_examples.py**

4.2.1.1 Direct evaluation using the Complexity class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `Complexity` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import Complexity
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = Complexity(
7     context=context,
8     params={
9         "normalise": True
10    }
11 )
12 scores = metric.run()

```

4.2.1.2 Automated evaluation using run_evaluation

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the `YAML` configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered "Complexity" metric, although this argument would not be necessary if `Complexity` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["Complexity"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset_name: "hydraulic"
  model_name: "IForest"
  xai_method_name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"
  attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: Complexity

```

```
params:  
  normalise: true
```

4.2.2 Sparseness

The Sparseness metric evaluates how concentrated the attribution magnitude is across the input features [16]. For each evaluated observation, the absolute attribution values are sorted, and the Gini index of the resulting attribution distribution is calculated.

An explanation that assigns most of its attribution magnitude to a small subset of features produces a high Sparseness score and is therefore considered more concise. Conversely, when the attribution magnitude is distributed relatively uniformly across many features, the score decreases. Thus, higher values indicate that fewer features account for most of the explanation, whereas values close to (0) indicate a more uniform attribution distribution.

The parameters used by this metric can be summarized as follows:

- **normalise** (bool): whether to normalise the attribution values before computing the Gini index. When this parameter is enabled, the default normalisation function provided by **Quantus** is used. (*Default: True*)

The absolute-value operation is always applied to the attribution values before computing the metric. This behaviour is defined by the underlying **Quantus** implementation and is fixed in the wrapper since the **abs** parameter is not exposed through the metric configuration. In addition, if all the attribution values supplied to the wrapper are negative, they are explicitly converted to their absolute values before being passed to **Quantus**.

The metric returns one Sparseness score for each evaluated observation. Higher values indicate that the attribution magnitude is concentrated on fewer features, while lower values indicate a more uniform distribution across the input features. The mean of the returned scores can be used as an aggregated measure of explanation sparseness for the evaluated observations.

The metric is implemented through the **Sparseness** class, which can be found in the module `xai_metrics.metrics.complexity`. It can be executed directly using this class (Section 4.2.2.1) or as part of the automated evaluation pipeline provided by `run_evaluation` (Section 4.2.2.2).

Both usage modes rely on the same **YAML** configuration file. The experiment evaluates explanations generated by **LIME** for an Isolation Forest model on the hydraulic dataset. Attribution normalisation is enabled before calculating the Gini-based Sparseness score.

Examples of both usage modes are available in the library under: `examples/specific_examples/Sparseness_examples.py`

4.2.2.1 Direct evaluation using the Sparseness class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric.

After constructing the context, the `Sparseness` class is instantiated directly using the desired parameters. Calling the `run()` method returns one consistency score for each evaluated observation, and the mean of these scores is calculated as an aggregated result.

```

1 from xai_metrics.config import ConfigController
2 from xai_metrics.metrics.faithfulness import Sparseness
3
4 config_path = "examples/specific_examples/config.yaml"
5 context, metadata = ConfigController(config=config_path).build_context()
6 metric = Sparseness(
7     context=context,
8     params={
9         "normalise": True
10    }
11 )
12 scores = metric.run()

```

4.2.2.2 Automated evaluation using run_evaluation

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the `YAML` configuration file. The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered `"Sparseness"` metric, although this argument would not be necessary if `Sparseness` is the only metric defined in the configuration.

```

1 from xai_metrics.runner import run_evaluation
2
3 config_path = "examples/specific_examples/config.yaml"
4 results = run_evaluation(
5     selected_metrics=["Sparseness"],
6     config=config_path,
7     report_output_dir=None
8 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset_name: "hydraulic"
  model_name: "IForest"
  xai_method_name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"
  attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: Sparseness

```

```
params:
  normalise: true
```

4.3 Sensitivity

This category includes metrics that evaluate how much an explanation changes when small modifications are introduced into the input data, the model parameters, or the explanation-generation process. For attribution-based explanations, these metrics analyse whether similar inputs produce similar attribution patterns and whether the explanation remains stable under minor or irrelevant perturbations.

Sensitivity metrics are particularly useful for identifying explanations that change excessively in response to small variations that do not substantially alter the model prediction. An explanation with low sensitivity is generally considered more stable, since minor perturbations produce only limited changes in the resulting attribution values. Conversely, high sensitivity indicates that small changes in the experimental conditions may lead to substantially different explanations.

However, sensitivity scores depend on the type, magnitude, and distribution of the perturbations applied during the evaluation. Perturbations that are too large may alter the semantic meaning of the input or change the model prediction, while perturbations that are too small may fail to reveal meaningful instability. The results may also depend on the number of perturbation samples and the distance function used to compare explanations. Consequently, sensitivity scores should be interpreted considering the perturbation strategy and the characteristics of the evaluated data and model.

4.3.1 Average Sensitivity

The Average Sensitivity metric evaluates how much an explanation changes, on average, when small random perturbations are applied to the corresponding input [17]. For each evaluated observation, several perturbed versions of the original input are generated, and their explanations are recomputed using an explanation function supplied by the user.

For each perturbation, the metric compared the new explanation with the original attribution vector. The sensitivity is calculated as the norm of the difference between both explanations divided by the norm of the original explanation. The final score is obtained by averaging these relative changes across all the generated perturbations.

A low Average Sensitivity score indicates that the explanation changes only slightly under small input perturbations and is therefore considered more stable. Conversely, a high score indicates that minor changes in the input produce substantially different explanations, suggesting greater sensitivity and lower robustness.

The parameters used by this metric can be summarized as follows:

- **nr_samples** (int): number of perturbed inputs generated for each evaluated observation. A larger number of samples provides a more extensive estimate of sensitivity but increases the computational cost because the explanation must be recomputed for every perturbed input. (*Default: 200*)
- **abs** (bool): whether to use the absolute values of the attributions before computing the metric. If all attribution values are negative and this parameter is enabled, their absolute values are used. Otherwise, the metric is skipped. (*Default: False*)

- **normalise** (bool): whether to normalise the original and perturbed attribution values before comparing their explanations. When enabled, the default normalisation function provided by **Quantus** is used. (*Default: False*)
- **lower_bound** (float): lower bound of the uniform noise used to generate the input perturbations. (*Default: 0.2*)
- **upper_bound** (float, optional): upper bound of the uniform noise used to generate the input perturbations. When no value is specified, the default behaviour defined by **Quantus** is used. (*Default: None*)

In addition to these parameters, the metric requires an external explanation function, provided through the **explain_func** argument. This function is necessary because the explanations must be recomputed after perturbing each input. It must be compatible with the interface expected by **Quantus** and use the following function signature:

```
1 def explain_func(model, inputs, targets=None, **kwargs):
2     ...
```

The **model** argument will contain the model being evaluated, **inputs** will contain the observations whose explanations must be generated, and **targets** optionally will identify the target outputs. The function must return a **NumPy array** containing the generated attribution vectors, with one row for each input observation. Additional arguments may be received through ****kwargs** for compatibility with the **Quantus** interface. If no explanation function is provided, the constructor raises a **ValueError**.

The implementation uses the default comparison, norm, normalisation, and perturbation functions provided by **Quantus**. By default, explanations are compared using their element-wise difference, while the numerator and denominator of the relative change are calculated using the Frobenius norm. The perturbed inputs are generated by adding uniform noise within the configured bounds.

Before computing the metric, the model is set to training mode, following the current implementation of the wrapper. The observations specified in the **MetricContext** are selected together with their target labels and original attribution values. The execution device stored in the context is also passed to **Quantus**.

The metric returns one Average Sensitivity score for each evaluated observation. Lower values indicate that the generated explanations are more stable under small random input perturbations, whereas higher values indicate greater variation between the original and perturbed explanations.

When all attribution values are negative, their treatment depends on the **abs** parameter. If **abs=True**, their absolute values are used and the evaluation continues. If **abs=False**, the metric raises a **MetricSkipped** exception and is not evaluated for that attribution configuration.

The metric is implemented through the **AvgSensitivity** class, placed in the module **xai_metrics.metrics.sensitivity**. It can be executed directly using this class (Section 4.3.1.2) or as part of the automated evaluation pipeline provided by **run_evaluation** (Section 4.3.1.3).

Both usage modes rely on the same **YAML** configuration file. The experiment evaluates explanations generated by **LIME** for an Isolation Forest model on the hydraulic dataset. For each observation, 200 perturbed inputs are generated using uniform noise with a lower bound of 0.2. The original signs of the attribution values are preserved, and attribution normalisation is disabled.

Examples of both usage modes are available in the library under: **examples/specific_examples/AvgSensitivity_examples.py**

The explanation function used by these examples is available under: **examples/specific_examples/make_explain_func.py**

4.3.1.1 LIME explanation function

The explanation function used in the examples is created by `make_lime_explain_func`. This factory receives the training data used as background information and creates a `LimeTabularExplainer`. It then returns a function that follows the interface required by Quantus.

The returned function checks whether the supplied model or its internal `model` attribute implements `predict_proba`. It converts the perturbed inputs into tabular representations, generates a LIME explanation for each observation, and returns the resulting attribution vectors as a NumPy array.

```

1 import numpy as np
2 import pandas as pd
3 import torch
4 from lime.lime_tabular import LimeTabularExplainer
5
6 def make_lime_explain_func(background):
7     columns = list(background.columns)
8     explainer = LimeTabularExplainer(
9         background.to_numpy(),
10        feature_names=columns,
11        mode="classification",
12        random_state=42,
13    )
14
15    def explain_func(model, inputs, targets=None, **kwargs):
16        detector = model
17        required_method = "predict_proba"
18        if hasattr(model, required_method):
19            detector = model
20        elif hasattr(model, "model") and hasattr(model.model, required_method):
21            detector = model.model
22        else:
23            raise AttributeError(f"No se encontro {required_method} en {type(model)} ni en
24                               model.model")
25
26        if isinstance(inputs, pd.DataFrame):
27            return inputs.loc[:, columns]
28        if isinstance(inputs, torch.Tensor):
29            values = inputs.detach().cpu().numpy()
30        else:
31            values = np.asarray(inputs)
32        if values.ndim == 1:
33            values = values.reshape(1, -1)
34        X_batch = pd.DataFrame(values, columns=columns)
35
36        attributions = []
37        for row in X_batch.to_numpy():
38            explanation = explainer.explain_instance(
39                data_row=row,
40                predict_fn=detector.predict_proba,
41                num_features=len(columns),
42            )
43            weights = np.zeros(len(columns), dtype=float)
44            for feature_idx, weight in explanation.as_map()[1]:
45                weights[feature_idx] = float(weight)
46            attributions.append(weights)

```

```

46         return np.asarray(attributions)
47
48     return explain_func

```

4.3.1.2 Direct evaluation using the AvgSensitivity class

In the direct evaluation mode, the experimental context is constructed from the `config.yaml` file using the `ConfigController` class. This context contains the model, test data, labels, attribution values, selected observations, and execution device required to calculate the metric. The training data are also loaded to create the LIME explanation function required to recompute the explanations of the perturbed inputs.

After constructing the context, the `AvgSensitivity` class is instantiated with the desired parameters and the explanation function returned by `make_lime_explain_func`. Calling the `run()` method returns one Average Sensitivity score for each evaluated observation, and the mean of these scores can be calculated as an aggregated result.

```

1  from make_explain_func import make_lime_explain_func
2  from xai_metrics.config import ConfigController
3  from xai_metrics.metrics.faithfulness import AvgSensitivity
4
5  X_train_path = "examples/data/hydraulic/X_train.csv"
6  X_train = pd.read_csv(X_train_path, index_col=0)
7
8  config_path = "examples/specific_examples/config.yaml"
9  context, metadata = ConfigController(config=config_path).build_context()
10 metric = AvgSensitivity(
11     context=context,
12     params={
13         "nr_samples": 200,
14         "abs": False,
15         "normalise": False,
16         "lower_bound": 0.2
17     },
18     explain_func = make_lime_explain_func(X_train)
19 )
20 scores = metric.run()

```

4.3.1.3 Automated evaluation using run_evaluation

The automated evaluation mode uses the `run_evaluation` function. The experimental context, the metrics to be executed, and their parameters are obtained from the YAML configuration file.

The explanation function cannot be defined directly in the YAML file. Therefore, it is created in Python and passed to `run_evaluation` through the `explain_func` dependency. The runner subsequently provides this function when instantiating metrics that require explanations to be recomputed.

The optional `selected_metrics` argument can be used to restrict the evaluation to a subset of the configured metrics. In this example, it selects only the registered "`AvgSensitivity`" metric, although this argument would not be necessary if `AvgSensitivity` is the only metric defined in the configuration.

```

1 from make_explain_func import make_lime_explain_func
2 from xai_metrics.runner import run_evaluation
3
4 X_train_path = "examples/data/hydraulic/X_train.csv"
5 X_train = pd.read_csv(X_train_path, index_col=0)
6
7 config_path = "examples/specific_examples/config.yaml"
8 results = run_evaluation(
9     selected_metrics=["AvgSensitivity"],
10    config=config_path,
11    report_output_dir=None,
12    explain_func = make_lime_explain_func(X_train)
13 )

```

The corresponding experimental configuration is defined as follows:

```

context:
  dataset_name: "hydraulic"
  model_name: "IForest"
  xai_method_name: "LIME"
  device: "cpu"
  model_path: "models/hydraulic/IForest/hydraulic_IForest_seed_0.pkl"
  X_test_path: "data/hydraulic/X_test.csv"
  y_test_path: "data/hydraulic/y_test.csv"
  attributions_path: "attributions/hydraulic/IForest/LIME/
    hydraulic_IForest_lime_attributions.csv"

metrics:
  - name: AvgSensitivity
    params:
      nr_samples: 200
      abs: false
      normalise: false
      lower_bound: 0.2

```

4.4 Robustness

This category includes metrics that evaluate the stability of explanations under small perturbations of the input, assuming that the corresponding model prediction remains approximately unchanged [15]. For attribution-based explanations, these metrics analyse whether similar observations produce similar attribution patterns and whether minor changes in the input lead to disproportionate variations in the generated explanations.

Robustness metrics are particularly useful for determining whether an explanation method produces stable and reliable results in local regions of the input space. A robust explanation should remain similar for observations that are close to each other and receive comparable model predictions. These metrics, therefore, make it possible to identify explanation methods whose outputs vary excessively despite limited changes in the input or model behaviour.

However, robustness scores may depend strongly on the perturbation strategy, the distance functions used to compare inputs and explanations, and the size of the neighborhood considered during the evaluations. Perturbations that are too large may alter the model prediction or generate

unrealistic observations, while excessively small perturbations may not reveal meaningful instability. The results may also be affected by differences in feature scales and by dependencies between input variables. Consequently, robustness scores should be interpreted considering the characteristics of the dataset, the model, and the experimental configuration.

Reporting bugs

Feel free to open an issue at Github if anything is not working as expected <https://github.com/kdislab/XAIMetrics/issues>. Merge request are also encouraged, it will be carefully reviewed and merged if everything is all right.

Bibliography

- [1] Emrullah ŞAHİN, Naciye Nur Arslan, and Durmuş Özdemir. Unlocking the black box: an in-depth review on interpretability, explainability, and reliability in deep learning, 1 2025.
- [2] In-On Wiratsin and Chaiyong Ragkhitwetsagul. Effectiveness of explainable artificial intelligence (xai) techniques for improving human trust in machine learning models: A systematic literature review. *IEEE Access*, 2025.
- [3] Logan Cummins, Alexander Sommers, Somayeh Bakhtiari Ramezani, Sudip Mittal, Joseph Jabour, Maria Seale, and Shahram Rahimi. Explainable predictive maintenance: A survey of current methods, challenges and opportunities. *IEEE Access*, 12:57574–57597, 2024.
- [4] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems 30*, pages 4765–4774. Curran Associates, Inc., 2017.
- [5] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should I trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Francisco, CA, USA, August 13-17, 2016*, pages 1135–1144, 2016.
- [6] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *International conference on machine learning*, pages 3319–3328. PMLR, 2017.
- [7] Alicja Gosiewska and Przemyslaw Biecek. Do not trust additive explanations. *arXiv preprint arXiv:1903.11420*, 2019.
- [8] Jianlong Zhou, Amir H. Gandomi, Fang Chen, and Andreas Holzinger. Evaluating the quality of machine learning explanations: A survey on methods and metrics, 3 2021.
- [9] Pedro Lopes, Eduardo Silva, Cristiana Braga, Tiago Oliveira, and Luís Rosado. Xai systems evaluation: A review of human and computer-centred methods, 10 2022.
- [10] Umang Bhatt, Adrian Weller, and José MF Moura. Evaluating and aggregating feature-based model explanations. *arXiv preprint arXiv:2005.00631*, 2020.
- [11] David Alvarez Melis and Tommi Jaakkola. Towards robust interpretability with self-explaining neural networks. *Advances in neural information processing systems*, 31, 2018.
- [12] Sanjoy Dasgupta, Nave Frost, and Michal Moshkovitz. Framework for evaluating faithfulness of local explanations. In *International Conference on Machine Learning*, pages 4794–4815. PMLR, 2022.
- [13] An-phi Nguyen and María Rodríguez Martínez. On quantitative aspects of model interpretability. *arXiv preprint arXiv:2007.07584*, 2020.

- [14] Marek Pawlicki, Aleksandra Pawlicka, Federica Uccello, Sebastian Szelest, Salvatore D’Antonio, Rafał Kozik, and Michał Choraś. Evaluating the necessity of the multiple metrics for assessing explainable ai: A critical examination. *Neurocomputing*, 602, 10 2024.
- [15] Anna Hedström, Leander Weber, Daniel Krakowczyk, Dilyara Bareeva, Franz Motzkus, Wojciech Samek, Sebastian Lapuschkin, and Marina M-C Höhne. Quantus: An explainable ai toolkit for responsible evaluation of neural network explanations and beyond. *Journal of Machine Learning Research*, 24(34):1–11, 2023.
- [16] Prasad Chalasani, Jiefeng Chen, Amrita Roy Chowdhury, Xi Wu, and Somesh Jha. Concise explanations of neural networks using adversarial training. In *International conference on machine learning*, pages 1383–1391. PMLR, 2020.
- [17] Chih-Kuan Yeh, Cheng-Yu Hsieh, Arun Suggala, David I Inouye, and Pradeep K Ravikumar. On the (in) fidelity and sensitivity of explanations. *Advances in neural information processing systems*, 32, 2019.